

AS7: Beta Estimation and Portfolio Analysis

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1 MLP

1.1 Model Architecture and Loss Function

The `NeuralBetaMLP` model is a two-layer feedforward neural network implemented in PyTorch. It consists of an input layer of size `in_dim`, a hidden layer of size `hidden_dim` with a selectable activation (`ReLU`, `Sigmoid`, `Tanh`, or `Linear`), and an output layer of size 1 producing the predicted beta. The model also includes a learnable intercept parameter α .

The loss function is defined as the root mean squared error (RMSE) between realized and predicted returns:

$$RMSE = \sqrt{E \left[(r_{t+1} - (\alpha + \hat{\beta} \cdot r_{m,t+1}))^2 \right]}.$$

1.2 Hyperparameter Search and Model Selection

A grid search was performed over the following hyperparameters:

- Lookback windows: 12, 24, 36
- Hidden units: 32, 64, 128
- Learning rates: 0.01, 0.001, 0.0003
- Activation functions: linear, sigmoid, tanh, ReLU

Each configuration was trained for 30 epochs with a batch size of 256. The model achieving the lowest validation RMSE was selected as the final configuration.

- **Selected parameters:**
 - Lookback window: 12
 - Hidden units: 128

- Learning rate: 0.0003
- Activation: ReLU

The final model was trained using these parameters for 30 epochs, and the training history was recorded.

1.3 Inference and Training Results

The model was trained for 30 epochs with a batch size of 256. Figure 1 shows the training and validation RMSE over epochs. The final training and validation RMSE after 30 epochs were 0.1139 and 0.0877, respectively. The test RMSE was 0.1280.

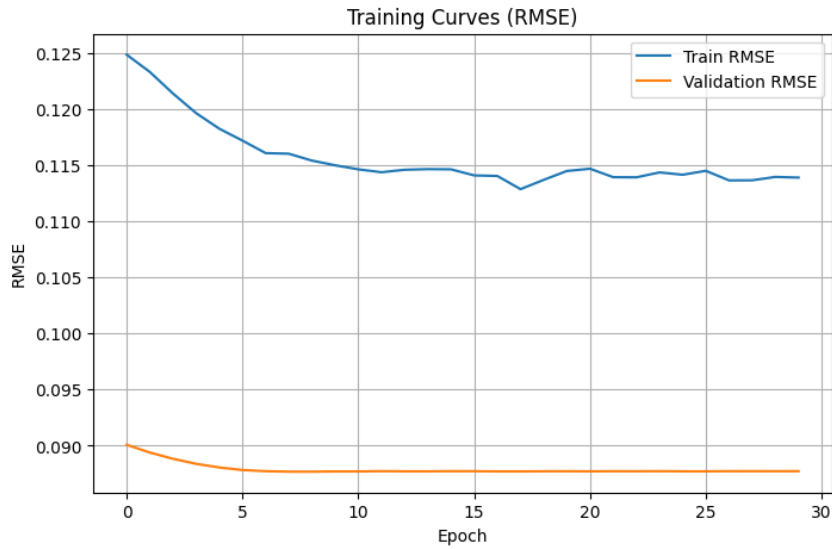


Figure 1: Training and Validation RMSE Curves over 30 Epochs

2 Beta Estimation and Portfolio Analysis

2.1 Industry-Level Beta Statistics Without Fama Factors

Table 1 presents the distribution of estimated betas across industries when only the market factor is included.

Table 1: Beta Statistics Without Fama Factors

Industry p99	N	Mean	Std	Skew.	Kurt.	Min	Max	p1	p5	p25	p50	p75	p95
Wholesale 1.982	539	1.095	0.421	-0.189	-0.868	0.325	2.211	0.353	0.395	0.752	1.183	1.418	1.673
Mining 1.782	648	0.844	0.407	0.543	-0.814	0.291	1.888	0.339	0.368	0.440	0.764	1.181	1.557
Manufacturing 1.767	601	0.912	0.376	0.220	-0.944	0.303	1.991	0.329	0.379	0.636	0.834	1.245	1.490
Finance 1.750	648	1.117	0.307	-0.153	-0.684	0.337	1.860	0.429	0.628	0.853	1.154	1.339	1.584
Transportation 1.895	360	1.300	0.252	0.081	1.890	0.604	2.574	0.692	0.829	1.158	1.302	1.453	1.663
Services 1.743	576	1.110	0.366	-0.421	-0.724	0.304	1.945	0.358	0.418	0.792	1.185	1.378	1.620
Construction 1.634	576	1.094	0.342	-0.529	-0.747	0.314	1.793	0.356	0.420	0.794	1.199	1.349	1.550
Retail 1.486	470	0.980	0.303	-0.408	-0.825	0.320	1.592	0.344	0.421	0.754	1.050	1.228	1.377

2.2 Industry-Level Beta Statistics With Fama Factors

Table 2: Beta Statistics With Fama Factors

Industry	N	Mean	Std	Skew.	Kurt.	Min	Max	p1	p5	p25	p50	p75	p95	p99
Wholesale	539	1.229	0.661	-0.045	-0.405	-0.498	2.924	-0.235	0.109	0.772	1.240	1.689	2.270	2.682
Mining	648	1.095	0.643	0.012	-0.433	-0.508	2.709	-0.282	0.063	0.657	1.098	1.550	2.194	2.552
Manufacturing	601	1.140	0.654	0.036	-0.364	-0.498	3.178	-0.292	0.059	0.669	1.168	1.575	2.233	2.606
Finance	648	1.281	0.620	-0.025	-0.468	-0.289	2.741	-0.030	0.256	0.808	1.300	1.692	2.282	2.674
Transportation	360	1.380	0.601	0.035	-0.461	-0.017	2.849	0.054	0.439	0.906	1.449	1.728	2.330	2.680
Services	576	1.258	0.637	-0.031	-0.429	-0.515	2.775	-0.100	0.220	0.776	1.282	1.684	2.289	2.646
Construction	576	1.261	0.639	-0.003	-0.377	-0.500	2.795	-0.107	0.205	0.782	1.254	1.675	2.299	2.671
Retail	470	1.198	0.630	0.016	-0.386	-0.502	2.682	-0.156	0.197	0.717	1.215	1.612	2.250	2.612

2.3 Portfolio Sorting by Beta Quintiles

Table 3: Equal-Weighted Returns by Beta Quintile

Quintile	Mean Beta	Equal-Weighted Return (EWRET)	N Stocks
1	0.3148	0.0138	883
2	0.8571	0.0142	884
3	1.2431	0.0205	883
4	1.5779	0.0093	884
5	2.1146	0.0063	884

Table 4: Value-Weighted Returns and Market Capitalization by Beta Quintile

Quintile	Mean Beta	VWRET	N Stocks	Total Market Cap (\$)
1	0.3148	0.0307	883	1.09×10^{10}
2	0.8571	0.0177	884	1.10×10^{10}
3	1.2431	0.0104	883	8.83×10^9
4	1.5779	0.0166	884	1.02×10^{10}
5	2.1146	0.0156	884	9.48×10^9

2.4 Q5 – Q1 Return Spreads

Table 5: Q5 – Q1 Return Spreads Across Beta Quintiles

Portfolio Type	Q5 Return	Q5 – Q1 Spread
Equal-Weighted (EW)	0.0063	-0.0074
Value-Weighted (VW)	0.0156	-0.0151

2.5 Interpretation: The Low-Beta Anomaly

Both equal-weighted and value-weighted portfolios show negative Q5–Q1 spreads, implying that high-beta stocks underperform low-beta stocks. This finding is consistent with the well-documented *low-beta anomaly*, where high-risk (high-beta) stocks yield lower realized returns than predicted by CAPM.

2.6 Comparison with Linear Regression Beta

Figures 2 and 3 together illustrate the differences between the traditional linear regression (LR) and machine learning (MLP) approaches to estimating betas. Figure 2 shows the cross-sectional distribution of betas by industry using LR over the full historical sample, highlighting considerable variation across sectors—particularly higher dispersion in Mining and Construction relative to Finance and Retail. In contrast, Figure 3, which presents the MLP-based beta dynamics for the period 2018–2023, reveals smoother and more stable trajectories across time and industries. This reflects the MLP model’s ability to regularize extreme values and capture nonlinear dependencies, while the LR model is more sensitive to historical shocks and outliers due to its linear specification and longer estimation window. Overall, the LR betas provide a broad long-run perspective of systematic risk, whereas the MLP betas emphasize recent, context-dependent stability and resilience in risk exposures.

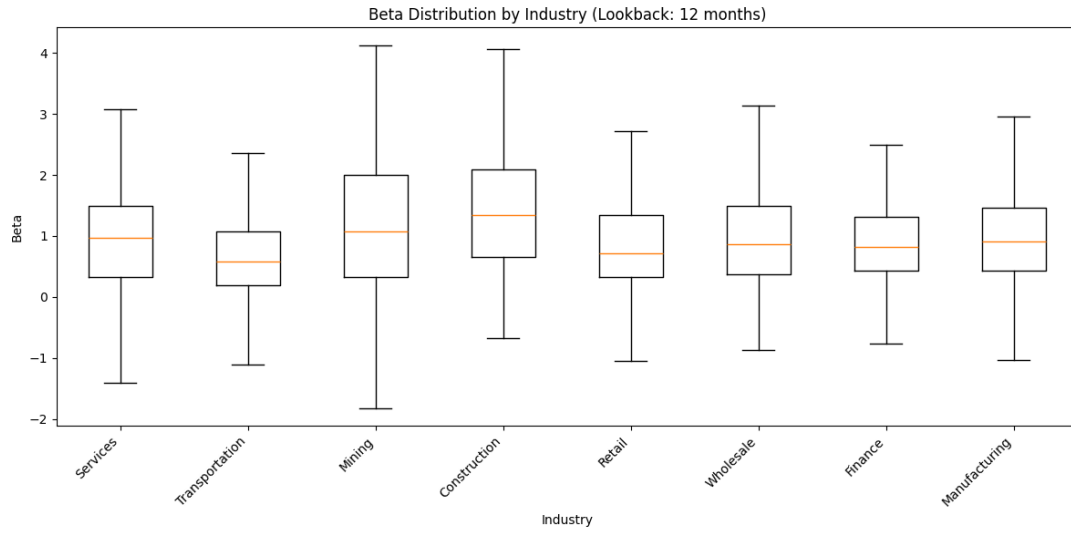


Figure 2: Beta Distribution by Industry (Linear Regression, full sample)

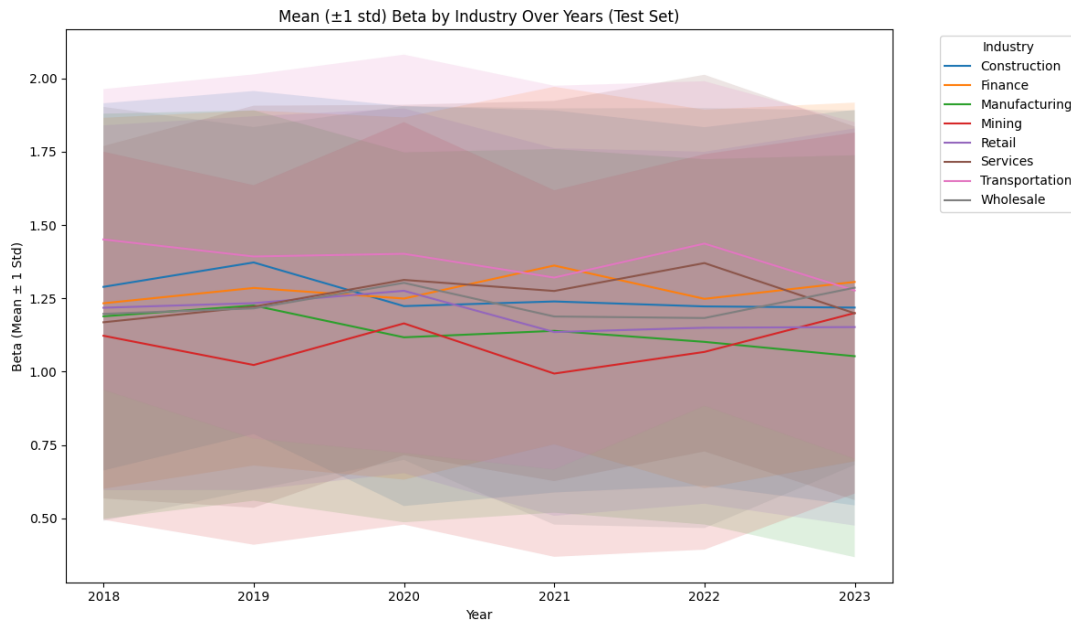


Figure 3: Mean (± 1 Std) Beta by Industry Over Time (MLP Model, 2018–2023)

3 Conclusion

The inclusion of Fama–French factors leads to higher and more dispersed betas. Portfolio sorting shows that high-beta portfolios earn lower average returns.